

- (b) Deduce from this result, and from Exercise 11, the Poisson integral representation formula from the Cauchy integral formula: If u is harmonic in the unit disc and continuous on its closure, then if $z = re^{i\theta}$ one has

$$u(z) = \frac{1}{2\pi} \int_0^{2\pi} P_r(\theta - \varphi) u(\varphi) d\varphi$$

where $P_r(\gamma)$ is the Poisson kernel for the unit disc given by

$$P_r(\gamma) = \frac{1 - r^2}{1 - 2r \cos \gamma + r^2}.$$

- 13.** Suppose f is an analytic function defined everywhere in $\mathbb{C}$ and such that for each $z_0 \in \mathbb{C}$ at least one coefficient in the expansion

$$f(z) = \sum_{n=0}^{\infty} c_n (z - z_0)^n$$

is equal to 0. Prove that f is a polynomial.

[Hint: Use the fact that $c_n n! = f^{(n)}(z_0)$ and use a countability argument.]

- 14.** Suppose that f is holomorphic in an open set containing the closed unit disc, except for a pole at z_0 on the unit circle. Show that if

$$\sum_{n=0}^{\infty} a_n z^n$$

denotes the power series expansion of f in the open unit disc, then

$$\lim_{n \rightarrow \infty} \frac{a_n}{a_{n+1}} = z_0.$$

- 15.** Suppose f is a non-vanishing continuous function on $\overline{\mathbb{D}}$ that is holomorphic in $\mathbb{D}$. Prove that if

$$|f(z)| = 1 \quad \text{whenever } |z| = 1,$$

then f is constant.

[Hint: Extend f to all of $\mathbb{C}$ by $f(z) = 1/\overline{f(1/\overline{z})}$ whenever $|z| > 1$, and argue as in the Schwarz reflection principle.]

7 Problems

- 1.** Here are some examples of analytic functions on the unit disc that cannot be extended analytically past the unit circle. The following definition is needed. Let

f be a function defined in the unit disc $\mathbb{D}$, with boundary circle C . A point w on C is said to be *regular* for f if there is an open neighborhood U of w and an analytic function g on U , so that $f = g$ on $\mathbb{D} \cap U$. A function f defined on $\mathbb{D}$ cannot be continued analytically past the unit circle if no point of C is regular for f .

(a) Let

$$f(z) = \sum_{n=0}^{\infty} z^{2^n} \quad \text{for } |z| < 1.$$

Notice that the radius of convergence of the above series is 1. Show that f cannot be continued analytically past the unit disc. [Hint: Suppose $\theta = 2\pi p/2^k$, where p and k are positive integers. Let $z = re^{i\theta}$; then $|f(re^{i\theta})| \rightarrow \infty$ as $r \rightarrow 1$.]

(b) * Fix $0 < \alpha < \infty$. Show that the analytic function f defined by

$$f(z) = \sum_{n=0}^{\infty} 2^{-n\alpha} z^{2^n} \quad \text{for } |z| < 1$$

extends continuously to the unit circle, but cannot be analytically continued past the unit circle. [Hint: There is a nowhere differentiable function lurking in the background. See Chapter 4 in Book I.]

2.* Let

$$F(z) = \sum_{n=1}^{\infty} d(n)z^n \quad \text{for } |z| < 1$$

where $d(n)$ denotes the number of divisors of n . Observe that the radius of convergence of this series is 1. Verify the identity

$$\sum_{n=1}^{\infty} d(n)z^n = \sum_{n=1}^{\infty} \frac{z^n}{1-z^n}.$$

Using this identity, show that if $z = r$ with $0 < r < 1$, then

$$|F(r)| \geq c \frac{1}{1-r} \log(1/(1-r))$$

as $r \rightarrow 1$. Similarly, if $\theta = 2\pi p/q$ where p and q are positive integers and $z = re^{i\theta}$, then

$$|F(re^{i\theta})| \geq c_{p/q} \frac{1}{1-r} \log(1/(1-r))$$

as $r \rightarrow 1$. Conclude that F cannot be continued analytically past the unit disc.

3. Morera's theorem states that if f is continuous in $\mathbb{C}$, and $\int_T f(z) dz = 0$ for all triangles T , then f is holomorphic in $\mathbb{C}$. Naturally, we may ask if the conclusion still holds if we replace triangles by other sets.

- (a) Suppose that f is continuous on $\mathbb{C}$, and

$$(16) \quad \int_C f(z) dz = 0$$

for every circle C . Prove that f is holomorphic.

- (b) More generally, let Γ be any toy contour, and $\mathcal{F}$ the collection of all translates and dilates of Γ . Show that if f is continuous on $\mathbb{C}$, and

$$\int_{\gamma} f(z) dz = 0 \quad \text{for all } \gamma \in \mathcal{F}$$

then f is holomorphic. In particular, Morera's theorem holds under the weaker assumption that $\int_T f(z) dz = 0$ for all equilateral triangles.

[Hint: As a first step, assume that f is twice real differentiable, and write $f(z) = f(z_0) + a(z - z_0) + b(\overline{z - z_0}) + O(|z - z_0|^2)$ for z near z_0 . Integrating this expansion over small circles around z_0 yields $\partial f / \partial \bar{z} = b = 0$ at z_0 . Alternatively, suppose only that f is differentiable and apply Green's theorem to conclude that the real and imaginary parts of f satisfy the Cauchy-Riemann equations.

In general, let $\varphi(w) = \varphi(x, y)$ (when $w = x + iy$) denote a smooth function with $0 \leq \varphi(w) \leq 1$, and $\int_{\mathbb{R}^2} \varphi(w) dV(w) = 1$, where $dV(w) = dx dy$, and $\int$ denotes the usual integral of a function of two variables in $\mathbb{R}^2$. For each $\epsilon > 0$, let $\varphi_{\epsilon}(z) = \epsilon^{-2} \varphi(\epsilon^{-1} z)$, as well as

$$f_{\epsilon}(z) = \int_{\mathbb{R}^2} f(z - w) \varphi_{\epsilon}(w) dV(w),$$

where the integral denotes the usual integral of functions of two variables, with $dV(w)$ the area element of $\mathbb{R}^2$. Then f_{ϵ} is smooth, satisfies condition (16), and $f_{\epsilon} \rightarrow f$ uniformly on any compact subset of $\mathbb{C}$.]

4. Prove the converse to Runge's theorem: if K is a compact set whose complement is not connected, then there exists a function f holomorphic in a neighborhood of K which cannot be approximated uniformly by polynomial on K .

[Hint: Pick a point z_0 in a bounded component of K^c , and let $f(z) = 1/(z - z_0)$. If f can be approximated uniformly by polynomials on K , show that there exists a polynomial p such that $|(z - z_0)p(z) - 1| < 1$. Use the maximum modulus principle (Chapter 3) to show that this inequality continues to hold for all z in the component of K^c that contains z_0 .]

- 5.* There exists an entire function F with the following “universal” property: given any entire function h , there is an increasing sequence $\{N_k\}_{k=1}^{\infty}$ of positive integers, so that

$$\lim_{n \rightarrow \infty} F(z + N_k) = h(z)$$

uniformly on every compact subset of $\mathbb{C}$.

- (a) Let $p_1, p_2, \dots$ denote an enumeration of the collection of polynomials whose coefficients have rational real and imaginary parts. Show that it suffices to find an entire function F and an increasing sequence $\{M_n\}$ of positive integers, such that

$$(17) \quad |F(z) - p_n(z - M_n)| < \frac{1}{n} \quad \text{whenever } z \in D_n,$$

where D_n denotes the disc centered at M_n and of radius n . [Hint: Given h entire, there exists a sequence $\{n_k\}$ such that $\lim_{k \rightarrow \infty} p_{n_k}(z) = h(z)$ uniformly on every compact subset of $\mathbb{C}$.]

- (b) Construct F satisfying (17) as an infinite series

$$F(z) = \sum_{n=1}^{\infty} u_n(z)$$

where $u_n(z) = p_n(z - M_n)e^{-c_n(z - M_n)^2}$, and the quantities $c_n > 0$ and $M_n > 0$ are chosen appropriately with $c_n \rightarrow 0$ and $M_n \rightarrow \infty$. [Hint: The function e^{-z^2} vanishes rapidly as $|z| \rightarrow \infty$ in the sectors $\{|\arg z| < \pi/4 - \delta\}$ and $\{|\pi - \arg z| < \pi/4 - \delta\}$.]

In the same spirit, there exists an alternate “universal” entire function G with the following property: given any entire function h , there is an increasing sequence $\{N_k\}_{k=1}^{\infty}$ of positive integers, so that

$$\lim_{k \rightarrow \infty} D^{N_k} G(z) = h(z)$$

uniformly on every compact subset of $\mathbb{C}$. Here $D^j G$ denotes the j^{th} (complex) derivative of G .

3 Meromorphic Functions and the Logarithm

One knows that the differential calculus, which has contributed so much to the progress of analysis, is founded on the consideration of differential coefficients, that is derivatives of functions. When one attributes an infinitesimal increase ϵ to the variable x , the function $f(x)$ of this variable undergoes in general an infinitesimal increase of which the first term is proportional to ϵ , and the finite coefficient of ϵ of this increase is what is called its differential coefficient... If considering the values of x where $f(x)$ becomes infinite, we add to one of these values designated by x_1 , the infinitesimal ϵ , and then develop $f(x_1 + \epsilon)$ in increasing power of the same quantity, the first terms of this development contain negative powers of ϵ ; one of these will be the product of $1/\epsilon$ with a finite coefficient, which we will call the residue of the function $f(x)$, relative to the particular value x_1 of the variable x . Residues of this kind present themselves naturally in several branches of algebraic and infinitesimal analysis. Their consideration furnish methods that can be simply used, that apply to a large number of diverse questions, and that give new formulae that would seem to be of interest to mathematicians...

A. L. Cauchy, 1826

There is a general principle in the theory, already implicit in Riemann's work, which states that analytic functions are in an essential way characterized by their singularities. That is to say, globally analytic functions are “effectively” determined by their zeros, and meromorphic functions by their zeros and poles. While these assertions cannot be formulated as precise general theorems, there are nevertheless significant instances where this principle applies.

We begin this chapter by considering singularities, in particular the different kind of point singularities (“isolated” singularities) that a holomorphic function can have. In order of increasing severity, these are:

- removable singularities